

## Module 4 Lesson 3

### Simple Interest (8.3)

---

## Objectives

In this lesson, we will learn how to

- calculate simple interest;
- use the future value formula.

## 1. Simple Interest

**Interest** is the dollar amount that we get paid for lending money or pay for borrowing money.

The amount of money that we deposit or borrow is called the **principal**.

The amount of interest depends on:

- the principal;
- the interest rate;
- the length of time the money is deposited or borrowed.

### Simple Interest

Simple interest is interest calculated only on the principal.

## 2. Simple Interest Formula

To calculate simple interest, we use the formula

$$\text{Interest} = \text{principal} \times \text{rate} \times \text{time}.$$

### Simple Interest Formula

$$I = Prt$$

where

$I$  = interest,

$P$  = principal,

$r$  = annual interest rate as a decimal,

and

$t$  = time in years.

### Important Reminder

The interest rate must be written as a decimal when using the formula.  
For example,

$$1.2\% = 0.012$$

and

$$8\% = 0.08.$$

## 3. Calculating Simple Interest for One Year

Example 1: You deposit \$2000 in a savings account at Hometown Bank. The account has an annual simple interest rate of 1.2%. Find the interest at the end of the first year.

### Solution

Use the simple interest formula.

$$I = Prt$$

Here,

$$P = 2000, \quad r = 1.2\% = 0.012, \quad t = 1.$$

Therefore,

$$I = (2000)(0.012)(1)$$

$$I = 24.$$

At the end of the first year, the interest is

$$\$24.$$

## 4. Calculating Simple Interest for More Than One Year

**Example 2:** A student took out a simple interest loan for \$1800 for 2 years at an annual rate of 8% to purchase a bicycle and a laptop. Find the interest on the loan.

### Solution

$$I = Prt$$

Here,

$$P = 1800, \quad r = 8\% = 0.08, \quad t = 2.$$

Therefore,

$$I = (1800)(0.08)(2)$$

$$I = 288.$$

The interest on the loan is

$$\$288.$$

## 5. Future Value: Principal Plus Interest

The **future value** is the total amount after interest is added to the principal.

In simple interest problems, the future value is

$$\text{Future value} = \text{principal} + \text{interest}.$$

Since

$$I = Prt,$$

we have

$$A = P + Prt.$$

Factoring out  $P$ , we get

$$A = P(1 + rt).$$

#### Future Value Formula

The future value,  $A$ , of  $P$  dollars at simple interest rate  $r$  for  $t$  years is

$$A = P(1 + rt).$$

Here,  $P$  is also called the present value.

## 6. Calculating Future Value

Example 3: A loan of \$1060 has been made at 6.5% simple interest for 3 months.  
Find the loan's future value.

### Solution

The variables are:

$$P = 1060,$$

$$r = 6.5\% = 0.065,$$

and

$$t = \frac{3}{12} = \frac{1}{4} = 0.25,$$

since 3 months is  $\frac{1}{4}$  of a year.

Use the future value formula:

$$A = P(1 + rt).$$

$$A = 1060[1 + (0.065)(0.25)]$$

$$A = 1060(1.01625)$$

$$A = 1077.225.$$

Rounded to the nearest cent, the loan's future value is

$$\$1077.23.$$

## 7. Determining a Simple Interest Rate

Sometimes we know the future value and need to find the interest rate.

Starting with

$$A = P(1 + rt),$$

we can solve for  $r$ .

### Solving for the Simple Interest Rate

$$A = P(1 + rt)$$

Divide both sides by  $P$ :

$$\frac{A}{P} = 1 + rt.$$

Subtract 1:

$$\frac{A}{P} - 1 = rt.$$

Divide by  $t$ :

$$r = \frac{\frac{A}{P} - 1}{t}.$$

Example 4: You borrow \$2500 from a friend and promise to pay back \$2655 in 6 months.  
What simple interest rate will you pay?

### Solution

Here,

$$A = 2655,$$

$$P = 2500,$$

and

$$t = \frac{6}{12} = \frac{1}{2} = 0.5.$$

Use

$$r = \frac{\frac{A}{P} - 1}{t}.$$

$$r = \frac{\frac{2655}{2500} - 1}{0.5}$$

$$r = \frac{1.062 - 1}{0.5}$$

$$r = \frac{0.062}{0.5}$$

$$r = 0.124.$$

Convert the decimal to a percent:

$$0.124 = 12.4\%.$$

You will pay a simple interest rate of

$$12.4\%.$$

## 8. Calculating a Present Value

Sometimes we know the future value we want to reach and need to find how much to invest now. This amount is the **present value**.

Starting with

$$A = P(1 + rt),$$

we can solve for  $P$ .

Present Value Formula

$$P = \frac{A}{1 + rt}$$

Example 5: You plan to save \$3600 for a trip to Europe in 2 years. You decide to purchase a certificate of deposit (CD) from your bank that pays a simple interest rate of 2.5%. How much must you put in this CD now in order to have \$3600 in 2 years?

### Solution

Here,

$$A = 3600, \quad r = 2.5\% = 0.025, \quad t = 2.$$

Use

$$P = \frac{A}{1 + rt}.$$

$$P = \frac{3600}{1 + (0.025)(2)}$$

$$P = \frac{3600}{1.05}$$

$$P \approx 3428.57.$$

Rounded to the nearest cent, you should put

$$\$3428.57$$

in the CD now to have \$3600 in 2 years.

## Summary

### Key Ideas

- Interest is the money paid for borrowing money or earned for lending money.
- Principal is the amount deposited or borrowed.
- Simple interest is calculated only on the principal.
- The simple interest formula is

$$I = Prt.$$

- The rate  $r$  must be written as a decimal.
- Time  $t$  must be written in years.
- The future value formula for simple interest is

$$A = P(1 + rt).$$

- The future value is the principal plus the interest.